[YOUR COLLEGE NAME]
DEPARTMENT OF [YOUR DEPARTMENT]

Literature Study

On

AI-Powered Smart Farming & Gardening Assistant

Submitted in partial fulfillment of the requirements for
9th Semester Mini Project

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Contents

Abstract

Agriculture and home gardening face persistent challenges in crop selection, disease management, water usage, and plant identification. Recent advances in Machine Learning (ML), Deep Learning (DL), Computer Vision, and the Internet of Things (IoT) have enabled data-driven decision support systems that address these problems. This literature study reviews existing research across five core areas relevant to the proposed *AI-Powered Smart Farming & Gardening Assistant*: (i) crop recommendation systems, (ii) deep learning-based plant/leaf disease detection, (iii) IoT-based smart irrigation systems, (iv) plant species identification using computer vision, and (v) integrated IoT-AI smart farming frameworks. The review identifies methodologies, datasets, and reported performance from existing works, followed by a comparative analysis and an explicit statement of research gaps that this project aims to address.

1 Introduction

Agriculture remains one of the largest economic sectors globally, yet farmers and home gardeners continue to rely heavily on manual observation and experience-based decisions for crop selection, irrigation, and disease management. The convergence of Artificial Intelligence (AI) and IoT technologies has opened new possibilities for precision agriculture and smart gardening, enabling real-time monitoring, predictive analytics, and automated decision support.

This literature study surveys prior academic and applied research relevant to the modules proposed in this project – Smart Farming and Smart Gardening – with a focus on four AI/ML components that form the technical core of the system: crop recommendation, plant disease detection, plant species identification, and IoT-driven irrigation automation. The objective is to understand existing approaches, their reported strengths and limitations, and to position the proposed system relative to the current state of the art.

2 Literature Review

2.1 Crop Recommendation Systems

Machine learning-based crop recommendation has been widely studied as a decision-support tool that maps soil and environmental parameters (nitrogen, phosphorus, potassium, pH, rainfall, temperature, humidity) to the most suitable crop. An IEEE-published system proposed a mobile-application-based crop recommender that additionally used GPS to localize recommendations to the farmer's region [1]. A broader decision-support study compared classical classifiers such as Random Forest, Support Vector Machines (SVM), K-Nearest Neighbour (KNN), and Naïve Bayes across multiple published works, concluding that ensemble methods such as Random Forest consistently outperform single-algorithm approaches [2].

More recent work has shifted toward ensemble boosting techniques combined with Explainable AI (XAI). One study trained a Gradient Boosting model on a standard crop-recommendation dataset (22 crop classes) and reported 99.27% accuracy, while integrating LIME-based explanations so that the reasoning behind a recommendation is interpretable rather than a ”black box” output [3]. A separate applied study corroborated similarly high classification accuracy using standard ML pipelines on comparable agricultural datasets [4].

Observation: Most existing crop recommendation systems are trained on a single, static, and often globally-sourced dataset (commonly the Kaggle ”Crop Recommendation” dataset), and few systems explain *why* a crop was recommended in farmer-accessible language – a gap directly relevant to this project's AI Assistant module.

2.2 Plant / Leaf Disease Detection using Deep Learning

Deep Convolutional Neural Networks (CNNs) have become the dominant approach for automated plant disease diagnosis from leaf images. In a foundational study, Mohanty et al. applied AlexNet and GoogLeNet architectures to identify 26 diseases across 14 crop species using the PlantVillage dataset, establishing the feasibility of CNN-based diagnosis at scale [5]. Sladojevic et al. used a Caffe-based deep learning framework to classify 13 plant diseases with 96.3% accuracy [6]. Ferentinos scaled this further, evaluating multiple CNN architectures (AlexNet variants, VGG) across 58 distinct plant diseases and real field images, achieving up to 99.53% accuracy with VGG-based models [7]. Geetharamani and Pandian proposed a lightweight, custom nine-layer CNN combined with data augmentation on the PlantVillage dataset, demonstrating that smaller, purpose-built architectures can outperform traditional ML while remaining computationally cheaper to train and deploy [8] – an important consideration for a resource-constrained academic project.

A 2022 systematic review synthesizing over 100 CNN-based disease-detection studies confirmed PlantVillage as the most widely used benchmark dataset and TensorFlow as the dominant development framework, while also noting that the majority of reported accuracies (often >95%) are obtained under controlled laboratory image conditions rather than cluttered, real-world field photographs [9].

Observation: A well-documented generalization gap exists between lab-condition accuracy and real-field performance. This motivates augmenting the PlantVillage dataset with real, self-captured leaf images during model training.

2.3 IoT-based Smart Irrigation Systems

IoT-enabled irrigation automation typically combines low-cost microcontrollers (ESP32/ESP8266, Arduino) with soil moisture, temperature, and humidity sensors to trigger watering without manual intervention. A comprehensive review of smart irrigation systems categorized existing approaches into ”suspended cycle” (fixed schedule) and ”water-on-demand” (thresholdtriggered) irrigation strategies, noting that threshold-based systems generally offer better water efficiency [10]. A systematic review of IoT solutions in smart farming organized the field into a standard four-layer IoT architecture – perception, network, processing, and application – and observed growing integration of cloud platforms and machine learning for irrigation decision-making beyond simple thresholds [11].

Applied studies report substantial efficiency gains: one field study measured up to a 70% reduction in water consumption compared to manual irrigation practices [12]. Velmurugan et al. extended threshold-based irrigation logic by additionally incorporating weather forecast data into the irrigation trigger decision, rather than relying on soil moisture alone [13].

Observation: Most deployed systems still rely on simple fixed-threshold logic. Combining live weather forecasts, soil moisture readings, and a trained ML model to predict irrigation *need* (rather than reacting to a fixed threshold) remains comparatively underexplored – a direction this project intends to pursue.

2.4 Plant Species Identification using Computer Vision

Large-scale plant identification has been demonstrated through transfer learning on deep CNN ensembles. Lasseck fine-tuned multiple state-of-the-art DCNNs to classify 10,000 plant species as part of the LifeCLEF 2017 / PlantCLEF challenge, achieving a mean reciprocal rank of 92% and top-5 accuracy of 96%, the best result among all participating teams [14]. In contrast, smaller-scope studies show that custom CNNs trained from scratch on narrower datasets can still achieve workable accuracy for constrained use cases – one such study trained a CNN from scratch on 10,413 images spanning 22 weed and crop species, achieving 86.2% classification accuracy under varied lighting and soil conditions [15]. A multi-division CNN (MD-CNN) approach improved fine-grained discrimination between visually similar species by partitioning leaf images into sub-regions before feature extraction [16].

Observation: State-of-the-art, large-vocabulary plant identification (10,000+ species) requires computational resources beyond the scope of a semester project. A transfer-learning approach (e.g., MobileNetV2 or EfficientNet fine-tuned on a curated set of common regional plants) is a more realistic and justifiable scope decision.

2.5 Integrated Smart Farming: IoT + AI/ML Survey Literature

Broader survey literature frames smart agriculture as the convergence of IoT sensing, cloud/edge computing, and ML/DL-based analytics across irrigation, disease detection, yield forecasting, and market decision support. A survey on smart farming technologies highlighted persistent challenges including sensor interoperability, connectivity limitations in rural areas, and the technical complexity of integrating heterogeneous data sources [17]. A more recent survey on IoT and AI/ML in smart agriculture reinforced that ML techniques used in practice range from simple linear regression and decision trees to more advanced neural network and deep learning approaches, depending on the sub-task [18]. An earlier systematic review of IoT solutions in smart farming (covering literature through 2020) similarly noted an increasing adoption of computer vision, big data, and machine learning layered on top of basic IoT sensing infrastructure [11].

Observation: Across surveys, a recurring gap is the lack of a single, unified, farmer/gardener-facing platform that integrates crop recommendation, disease detection, irrigation automation, and plant identification under one system with human-interpretable explanations – rather than isolated, single-purpose research prototypes.

3 Comparative Analysis of Reviewed Literature

Study	Technique	Key Result	Limitation
Mohanty et al. [5]	AlexNet, GoogLeNet CNN	26 diseases, 14 crop species identified via PlantVillage	Lab-condition images only
Sladojevic et al. [6]	Caffe-based CNN	96.3% accuracy, 13 diseases	Limited disease diversity
Ferentinos [7]	AlexNetOWTBn, VGG	58 diseases, up to 99.53% accuracy	Field-image generalization not fully addressed
Geetharamani & Pandian [8]	Custom 9-layer CNN + augmentation	Outperforms traditional ML on PlantVillage	Small custom architecture, single dataset
Springer XAI Study [3]	Gradient Boosting + LIME	99.27% crop recommendation accuracy, explainable output	Single static dataset (22 crops)
Lasseck [14]	Ensembled DCNNs (transfer learning)	10,000 species, 96% top-5 accuracy	Extremely high compute/data requirement
ScienceDirect Species CNN [15]	CNN trained from scratch	86.2% accuracy, 22 species	Lower accuracy, small species set
Velmurugan et al. [13]	ESP-based IoT + weather data	Weather-integrated irrigation triggering	Rule-based, not ML-driven
Ayaz et al. [11]	Systematic review (IoT architecture)	4-layer IoT reference architecture	Review only, no novel system

Study	Technique	Key Result	Limitation
Idoje [17]	Survey	Identifies smart-farming adoption challenges	Survey only, no implementation

4 Research Gap

Based on the reviewed literature, the following gaps are identified:

1. **Fragmentation:** Existing research largely addresses crop recommendation, disease detection, irrigation, and plant identification as *separate, isolated* systems rather than an integrated platform.
2. **Explainability:** Very few crop recommendation or disease detection systems explain predictions in farmer/gardener-accessible language; most function as opaque classifiers.
3. **Field-condition robustness:** Disease detection models trained purely on lab-condition datasets (e.g., PlantVillage) show a documented drop in accuracy on real, cluttered field images.
4. **Irrigation intelligence:** Most IoT irrigation systems rely on fixed soil-moisture thresholds rather than a learned model that combines weather forecasts, soil data, and crop-specific water needs.
5. **Dual-domain coverage:** No reviewed system simultaneously serves both professional farming and home-gardening use cases with a shared underlying AI pipeline.

The proposed *AI-Powered Smart Farming & Gardening Assistant* directly addresses these gaps by combining crop recommendation, plant/leaf disease detection (shared model across both modules), IoT-driven irrigation with weather integration, and plant species identification into a single, explainable, dual-purpose platform.

5 Conclusion

The reviewed literature confirms that machine learning and deep learning techniques have matured sufficiently to support reliable crop recommendation, plant disease detection, and species identification, while IoT-based sensing has proven effective for automating irrigation and reducing water waste. However, existing work remains largely fragmented across single-purpose systems, with limited attention to explainability, field robustness, and unified farmer-gardener platforms. This project is positioned to address these identified gaps through an integrated, modular, and explainable AI-driven system.

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